

README

llm-alias-health

Revision: Last modified:
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PWU- of the LLMsVerifier incorporation (see `docs/research/llmsverifier_incorporation_20260707/ANALYSIS_AND_PLAN.md` , PART D + PART F, in the ATMOSphere-Android- consuming repo) – the permanent, project-agnostic auto-select **health engine** for Claude-Code-CLI aliases (native accounts, provider-router routes via `ccr` , and provider-native routes), replacing the previous *reactive* cooldown-only rotation with a *proactive* out-of-band health probe.

What it does

For every alias in a YAML config, `llm-alias-health` :

- . **Execs** the LLMsVerifier `claude-alias-probe` bridge binary (`$ . . (C) exec-decoupling` – this util never imports LLMsVerifier as a Go dependency; the bridge path is injected via `llmsverifier_bin` in the config, never hardcoded).
- . Classifies the result into the closed-set status vocabulary `{ok, rate_limited, quota_exceeded, failed, unknown}` – `unknown` is reserved for cases where the bridge itself could not be run or its output could not be parsed (missing binary, crash, timeout, corrupted stdout); it is **never** conflated with a genuine `ok` .

- Writes one evidence JSON file per alias
(`ab_pass_with_evidence` style artefact, \$ `ab_pass_with_evidence`) plus
an append-only `events.jsonl` real-time event trail
(\$ `ab_pass_with_evidence`).
- Atomically rewrites a native-preferred-ranked `health.json`
snapshot a multitrack consumer can tail.

Config schema

```
llmsverifier_bin: /path/to/claude-alias-probe # required
timeout_seconds: 60 # optional, default 60
sentinel_prefix: "LLM_ALIAS_HEALTH_" # optional
aliases:
  - alias: claude1
    kind: native # native | provider-native | provider-router
    config_dir: /home/user/.claude-claude1
    model: claude-opus-4-8 # optional
  - alias: zai-glm-router
    kind: provider-router
    base_url: http://127.0.0.1:3456
  - alias: some-provider
    kind: provider-native
    base_url: https://api.example.com
    auth_token_env: SOME_PROVIDER_TOKEN # NAME of the env var holding the secret
```

Usage

```
go build -o bin/llm-alias-health ./cmd/llm-alias-health
./bin/llm-alias-health \
  -config health_config.yaml \
  -health-out health.json \
  -events-out events.jsonl \
  -evidence-dir qa-results/llmsverifier_pwu3/evidence
```

Exit code `2` on config/usage error; otherwise `0` (a per-alias probe failure is reflected in `health.json`, it is never a reason for the whole sweep to fail).

Testing

```
go test -p 2 ./... - unit (config parse, status mapping,
ranking, atomic write) + a full-sweep end-to-end run against
a controllable stub bridge (testdata/stubbridge) + stress
(      -iteration latency percentiles,      -way concurrency)
+ chaos (corrupted/empty/missing-binary /hung-subprocess/
unwritable-evidence-dir/pre-corrupted-events-file - all
degrade honestly, never a fabricated ok ).
```

```
go test -p 2 -tags integration ./... drives the real
LLMsVerifier bridge via LLM_ALIAS_HEALTH_TEST_BRIDGE_BIN=<built-
binary-path> (one real claude -p call via ccr, quota-
minimal).
```

Status

Go code + full test suite (unit/stress/chaos/integration) complete. Four-format doc export (HTML/PDF/DOCX per `$ . . . /$ . . .`) and wiring into the shared `meta_test_false_positive_proof.sh` paired- mutation harness are tracked as PWU- follow-up per the incorporation plan's own phasing, not part of this PWU's Go-code deliverable.